WORKED SOLUTION

STEP I 1987, Question 10

Mechanics • Hooke's Law

STEP I 1987, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s1-q10

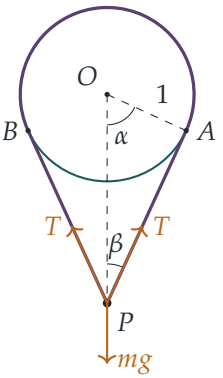
The key idea

The particle drags the bottom of the band down off the cylinder, so the band has two parts: an arc that still hugs the cylinder over the top, and two straight pieces that leave the cylinder *along tangents* and meet at the particle. The stretched length is then pure geometry: right-angled triangles give the straight pieces and the angle subtended at the axis. Because the cylinder is smooth, the tension is the same all the way round, and resolving at the particle links that tension to mg; Hooke's law then gives λ.

Solution

The model

We model the band as a light elastic string of natural length 2π obeying Hooke's law, the cylinder as smooth, and the particle as a point mass. We work in a vertical cross-section: the cylinder is a unit circle with centre O on the axis, and the particle is at P, a distance x vertically below O. For the band to leave the cylinder at all we need $x > 1$.



By symmetry about the vertical through O, the band leaves the cylinder at two points

A and B, one on each side. A taut string that comes off a smooth curve leaves it along the tangent, so PA is tangent to the circle at A, which means $OA \perp AP$. Let $\alpha = \angle AOP$ and $\beta = \angle APO$.

The stretched length

Triangle OAP has a right angle at A, with $OA = 1$ and hypotenuse $OP = x$. Hence

$$AP = \sqrt{x^2 - 1}, \qquad \cos\alpha = \frac{1}{x}, \qquad \sin\beta = \frac{1}{x}.$$

The band is in contact with the cylinder everywhere except between A and B through the bottom, an angle of 2α at O. So the arc in contact subtends $2\pi - 2\alpha$, and on a unit circle its length is $2\pi - 2\alpha$. Adding the two straight pieces, the stretched length is

$$\boxed{L = 2\sqrt{x^2 - 1} + 2\pi - 2\arccos\frac{1}{x}}$$

where $\arccos(1/x)$ lies between 0 and $\frac{\pi}{2}$, since $x > 1$. The extension is therefore

$$L - 2\pi = 2\sqrt{x^2 - 1} - 2\arccos\frac{1}{x}.$$

Ask yourself The band is curved round the cylinder. Why is the tension the same in the straight pieces as in the arc?

On a small piece of band in contact with a smooth surface, the only force from the cylinder is normal to the surface, so there is nothing to balance a change in tension along the band. Hence the tension T is constant along the whole band, and Hooke's law applies to the band as a whole:

$$T = \frac{\lambda\,(L - 2\pi)}{2\pi}.$$

(If the cylinder were rough, the tension could vary round the arc, and this step would fail.)

Equilibrium of the particle

The particle is acted on by its weight mg and by tensions T along PA and PB, each at angle β to the upward vertical. The horizontal components cancel by symmetry. Resolving vertically,

$$2T\cos\beta = mg, \qquad \text{with} \qquad \cos\beta = \frac{\sqrt{x^2 - 1}}{x},$$

so

$$T = \frac{mgx}{2\sqrt{x^2 - 1}}.$$

The case $x = 2$

Now $\cos\alpha = \frac{1}{2}$, so $\alpha = \frac{\pi}{3}$, and $\sqrt{x^2-1} = \sqrt{3}$. The stretched length is

$$L = 2\sqrt{3} + 2\pi - \frac{2\pi}{3} = 2\sqrt{3} + \frac{4\pi}{3}, \qquad \text{so} \qquad L - 2\pi = 2\sqrt{3} - \frac{2\pi}{3}.$$

The tension is $T = \dfrac{2mg}{2\sqrt{3}} = \dfrac{mg}{\sqrt{3}}$. Hooke's law gives

$$\lambda = \frac{2\pi T}{L - 2\pi} = \frac{2\pi mg/\sqrt{3}}{2\sqrt{3} - 2\pi/3} = \frac{2\pi mg}{6 - 2\pi/\sqrt{3}} = \frac{\pi mg}{3 - \pi/\sqrt{3}},$$

and multiplying top and bottom by $\sqrt{3}$,

$$\boxed{\lambda = \frac{\sqrt{3}\,\pi\, mg}{3\sqrt{3} - \pi} \approx 2.65\, mg.}$$

Checks

- As $x \to 1^+$, $\arccos(1/x) \to 0$ and $\sqrt{x^2-1} \to 0$, so $L \to 2\pi$: the band just touches the bottom of the cylinder and is unstretched, as it should be. Meanwhile $T \to \infty$, which matches the fact that two nearly horizontal strings can only hold up a weight with a huge tension.
- At $x = 2$, $L = 2\sqrt{3} + 4\pi/3 \approx 7.653$, a little more than $2\pi \approx 6.283$, and λ has the units of a force, like mg.
- Independently, we minimised the total potential energy $-mgx + \lambda(L - 2\pi)^2/(4\pi)$ numerically with $\lambda \approx 2.6484\, mg$; the minimum is at $x = 2$, which confirms both the geometry and the tension.

Common slips

- Taking the straight pieces from P to the ends of the horizontal diameter. The band leaves along tangents, so it comes off *below* the level of the axis, at angle $\alpha = \arccos(1/x)$ from the downward vertical.
- Using the extension of one straight piece in Hooke's law. The tension is uniform, so it's the extension of the whole band, $L - 2\pi$, divided by the whole natural length 2π.

Beyond STEP

The claim that tension is constant round a smooth surface is the frictionless case of the capstan equation: with coefficient of friction μ, the tension can change by a factor of up to $e^{\mu\theta}$ around an angle θ. That is why a rope wrapped a few times round a

post can hold a ship. The energy check above is also a first taste of the variational view of mechanics, in which equilibria are stationary points of potential energy. For a first-year treatment of forces, friction and potential energy in this spirit, read Kleppner and Kolenkow, *An Introduction to Mechanics*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.